D RAGHURAM

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EDUCATION

SRM Institute of Science and Technology **Chennai, India**

B.Tech in Computer Science and Engineering (Specialization: Big Data Analytics) 2025 – 2029 (Expected)

- **CGPA:** 8.54 / 10.0
- **Relevant Coursework:** Data Structures & Algorithms (DSA), Database Management Systems (DBMS), Operating Systems (OS), Object-Oriented Programming (OOP), Prompt Engineering.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, SQL, C, C++
- **Data Analysis & Visualization:** NumPy, Pandas, Matplotlib, Seaborn
- **Backend, Web & Databases:** Django, JDBC, REST APIs, Java Swing, Cryptographic Hashing, HTML, CSS
- **Core Concepts & Developer Tools:** Object-Oriented Programming (OOP), Design Patterns, Multithreading, Git, GitHub, Linux

ENGINEERING PROJECTS

MealSynx — *Intelligent Campus Food Demand, Waste & Resource Optimization Platform* **github.com/D-Raghuram-f4a/mealsynx**

Multi-tier campus dining optimization system integrating real-time demand forecasting, dynamic multi-canteen load balancing, and closed-loop waste analytics.

- **Multi-Tier System Architecture:** Engineered core backend services in Java 21 with an embedded REST API, unifying 10 end-to-end modules across demand forecasting, dynamic multi-canteen load balancing, and closed-loop waste analytics.
- **3NF Relational Database Design:** Architected a normalized 12-table schema in MySQL 8.4 integrated via JDBC to automate recipe Bill of Materials (BOM) ingredient explosion and real-time digital wallet/UPI transaction persistence.
- **Extensible Simulation Engine:** Implemented the Strategy design pattern to execute "What-If" demand simulations incorporating contextual variables such as weather shifts and campus event schedules.
- **Streaming & Big Data Pipeline:** Built an event-driven ingestion stream using Apache Kafka paired with batch demand modeling in Apache Spark to minimize food overproduction and peak-hour stockouts.
- **High-Concurrency Reliability:** Developed modular service layers and automated input validation in Java 21 to ensure ACID-compliant transaction handling and low-latency query execution across concurrent canteen terminals.